Quantitative Investing: Forecasting, Market Models, and Portfolio Construction

SMM921: Trading and Microstructure

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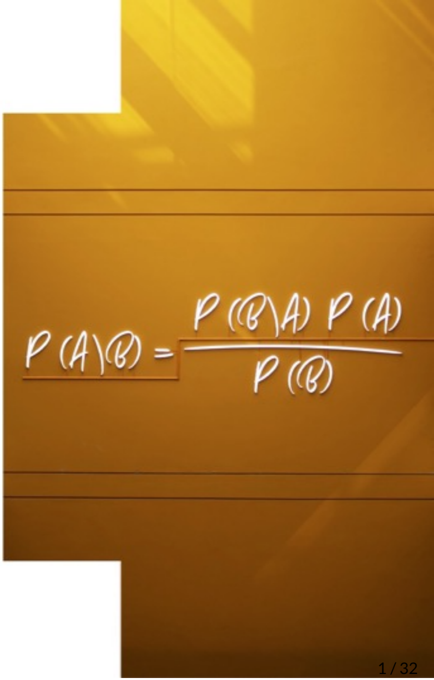

Recap

The key points from our previous session:

- Being a good active portfolio manager requires an ability to:
  - Forecast returns accurately
  - Measure risk
  - Assess investor risk aversion
  - Predict transaction costs
- Good portfolio managers are characterised by high information ratios (IR)
- Maximising the value added from a portfolio manager is equivalent to choosing the manager with the greatest IR

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Reading

In this section we focus on understanding how IRs are determined, how to construct alphas and the models managers often use to measure risk.

Look at:

- Ang, Chs. 6, 7 and 10
- Grinold and Kahn, Chs. 6 and 10
- Grinold and Kahn, “Advances in Active Portfolio Management”, Chs. 21 and 23
- Kinlaw, Kritzman and Turkington, Ch. 6

Active management: the Fundamental Law I

Grinold (1989) provides a way in which to decompose information ratios which provides some insights into the determinants of superior performance.

The Fundamental Law of active management

Information ratios are determined by two factors:

- Breadth (BR): the number of independent forecasts made in a given period
- Information coefficient (IC): a measure of forecasting accuracy – the correlation between forecasts and outcomes

In particular, the information ratio (IR) can be written as:

$$\mathrm{IR} = \mathrm{IC} \times \sqrt{\mathrm{BR}}$$

Active management: the Fundamental Law II

Implications: the fundamental law tells us that

- Information ratios are linearly increasing in forecast quality
- Information ratios increase with the square root of breadth, which increases if
  - we increase the number of different signals that we use to forecast returns
  - we increase the size of our investment universe

Implications for value added: substituting the fundamental law into the expression for optimal value added we get

$$\mathrm{VA}^* = \frac{\mathrm{IR}^2}{4\lambda} = \frac{\mathrm{IC}^2 \times \mathrm{BR}}{4\lambda}$$

So value added is linear in BR and increases with the square of IC.

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Fundamental Law: examples I

Assume you want a portfolio/manager with an IR of 0.5. Both of the following cases will deliver that.

1. A cross-sectional factor strategy (e.g. momentum):

- BR: trading the S&P 500, so we have (in principle) 500 potential bets at a point in time
- IC: the correlation between forecasts from your factor and actual subsequent (residual) stock returns is 0.025
- IR $= \mathrm{IC} \times \sqrt{\mathrm{BR}} = 0.025 \times \sqrt{500} > 0.5$

Intuition: you make money because, despite the fact you are not a very good forecaster, you make lots of bets.

Fundamental Law: examples II

2. A market timing strategy (forecasting the FTSE 100 Index):

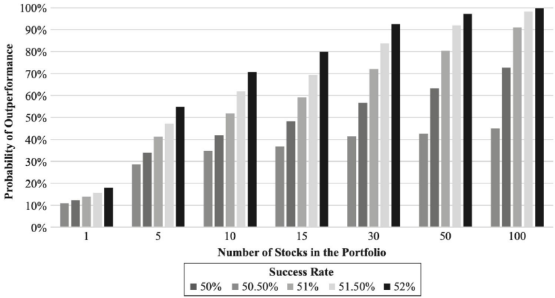
- BR: trading the FTSE 100 index once every quarter (i.e. 4 bets a year on 1 asset, so breadth is 4)
- IC: the correlation between forecasts from your model and actual subsequent (residual) FTSE 100 returns is 0.25
- IR $= \mathrm{IC} \times \sqrt{\mathrm{BR}} = 0.25 \times \sqrt{4} = 0.5$

Intuition: you make money because, despite the fact that you make very few bets, you are a good forecaster.

Summary: you maximise your chances of making money by betting well (high IC) *and* betting frequently (high BR).

Active skill and number of bets

Probability of outperforming the benchmark increases as success rate (IC) and portfolio size (BR) increase



Notes: Based on quarterly Russell 3000 Index constituents return data from January 1987 through December 2017.
Sources: Tidmore, Kinniry, Renzi-Ricci and Cilla, 2019, "How to Increase the Odds of Owning the Few Stocks that Drive Returns", *The Journal of Investing*, 29(1), 43-60.

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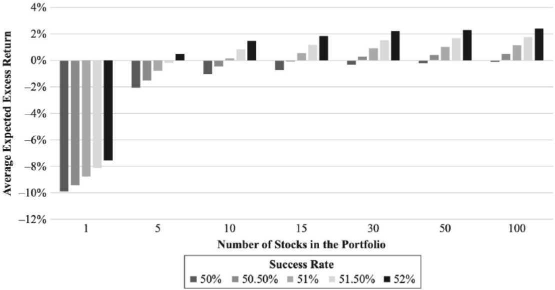
Active skill and number of bets

Average expected excess return rises as success rate (IC) and portfolio size (BR) increase



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Forecasting returns

We have encountered some important concepts related to return forecasting, but I have given you no way to think about them practically:

- Alpha: a forecast of (residual) returns
- IC: the correlation between forecast returns and actual subsequent returns

In this section we will cover ways to compute alphas and estimate ICs from:

1. Time-series models
2. Cross-sectional models

Forecasting via time-series regression I

A simple setting: 1 asset and 1 forecast. Residual returns (alpha) are denoted r_t^* and your predicting variable (e.g. earnings or dividend/price ratios) is s_t. Run the regression:

$$r_t^* \;=\; a + b\, s_t + e_t$$

What are the optimal OLS estimates of the parameters?

$$\widehat{b} \;=\; \frac{\mathrm{Cov}(r_t^*,\, s_t)}{\mathrm{Var}(s_t)} \;=\; \rho_{r^*,s}\,\frac{\sigma_{r^*}}{\sigma_s}\,, \qquad \widehat{a} \;=\; \overline{r^*} - \widehat{b}\,\overline{s} \;=\; -\widehat{b}\,\overline{s}$$

Note that the mean residual return is (by definition) zero. Then define

$$\mathrm{IC} \;=\; \rho_{r^*,s} \;=\; \mathrm{corr}(r_t^*,\, s_t)$$

Forecasting via time-series regression II

We can then write

$$\widehat{r^*}_t \;=\; \widehat{a} + \widehat{b}\, s_t \;=\; \mathrm{IC} \cdot \sigma_{r^*} \cdot \frac{s_t - \overline{s}}{\sigma_s}$$

Finally, our best guess of the dependent variable from a regression, given the value of the independent variable, is

$$E\left[r^*_t \mid s_t\right] \;=\; \mathrm{IC} \cdot \sigma_{r^*} \cdot z_{s,t}\,, \qquad z_{s,t} \equiv \frac{s_t - \overline{s}}{\sigma_s}$$

So your best forecast of the residual return is the product of forecasting skill, residual return volatility, and the standardised value of the signal.

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Forecasting via time-series regression III

- In practice, time series models tend to be more complex:

$$r^*_{t+1} \;=\; c_0 + c_1 s_t + c_2 k_t + r^*_t + e_{t+1}$$

- For equities, this approach is not very common (factor models tend to be preferred), but in multi-asset portfolio construction a vector based approach (VAR) is very often used:

CORE MODULE
Country-specific VAR with correlated shocks

$\longrightarrow$

ATTRIBUTION MODULE
Asset classes' risk and return attribution

$$\mathbf{y}_t = \mathbf{y}_{t-1} + \mathbf{e}_t \;\longrightarrow\; \widehat{\mathbf{y}}_{t+1}$$

$$\mathbf{r}_t = f(\mathbf{y}_t, \epsilon_t) \;\longrightarrow\; \widehat{\mathbf{r}}_{t+1} = f(\widehat{\mathbf{y}}_{t+1}, \epsilon_t)$$

Forecasting via time-series regression IV

Interpretation of the IC:

- Now you can see explicitly that the IC is the correlation between the forecasting variable and the return.
- Your best forecast of the residual return is equal to the product of:
  - Forecasting skill (IC)
  - The volatility of the residual return
  - A standardised version of the forecasting variable

Intuitively, this makes a lot of sense: the forecast will be large and positive if

1. the signal is well above its mean
2. you are very confident in the quality of the signal variable
3. the range of possible values of the residual return is large

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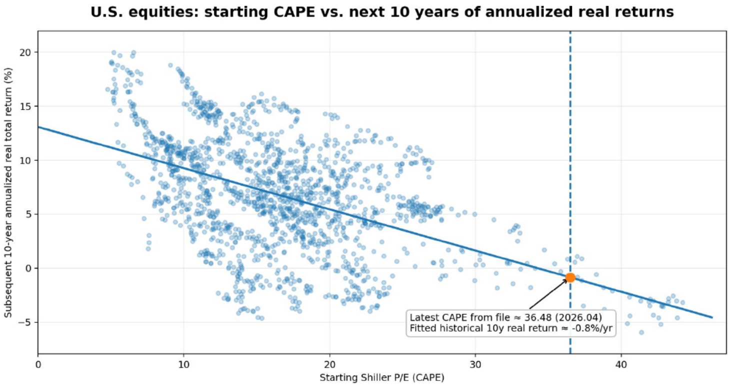
Are equity returns predictable?



Notes: Monthly observations from January 1881 to April 2026.
Sources: Shiller dataset: https://shillerdata.com

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Constructing alphas: example I

For a buy/sell list: often you are given such a list by an investment team.

- Convert recommendations to numeric signals: buy $= +1$ and sell $= -1$.
  - As long as the buy and sell lists are equal in length, these are already standardised signals.
- Select an IC. Here we choose 0.05.
- Then just compute alphas as:

$$\widehat{\alpha}_i \; = \; \text{signal}_i \times \text{IC} \times \sigma_{i,\text{resid}}$$

Constructing alphas: example II

Worked example:

Stock	View	Signal	Risk	Alpha
BARC	Buy	$+1$	30%	$+1.5\%$
GSK	Buy	$+1$	26%	$+1.3\%$
MKS	Sell	-1	16%	-0.8%
VOD	Sell	-1	20%	-1.0%

Forecasting via cross-sectional regression

A cross-sectional approach forecasts returns across stocks *at a point in time*, rather than through time for the market or one stock:

$$r^*_{i,t+1} \;=\; a + b\, s_{i,t} + e_{i,t+1}$$

So instead of asking "What will equity returns be next month?", you ask "Which stocks will outperform other stocks next month?". In practical active management, this becomes:

1. Build factor scores for each stock
2. Combine them into an alpha score
3. Rank stocks by expected excess return
4. Form long-short or overweight-underweight portfolios

Often preferred for active equity because managers usually care more about relative performance across stocks than about forecasting the whole market level.

Risk models

The other basic input to an optimization problem is the covariance matrix of returns for your stock universe. How is this typically estimated?

- **Historical sample covariance matrix**
  - Often a suboptimal choice, despite its continued popularity
  - Prone to measurement error
  - Lacks sufficient structure
- **Factor model**
  - Imposes structure on how covariances are determined
  - Can produce more stable and robust portfolios

Factor model: structure I

The standard setup of a factor structure for returns is as follows:

$$r_{i,t} = \alpha_i + \sum_{k=1}^{K} \beta_{i,k} F_{k,t} + \varepsilon_{i,t} \tag{1}$$

Returns are decomposed into:

- A constant term
- Variation caused by exposures (the β coefficients) to a set of K systematic factor returns (the Fs) that affect all stocks
- An idiosyncratic return component (the ε term), which has zero mean and is typically uncorrelated across stocks

Assuming the idiosyncratic risk components are uncorrelated across stocks implies that our K factors account for all cross-stock return comovement.

Factor model: structure II

Simple example: the CAPM/market model. We have

$$r_{i,t} \;=\; \alpha_i + \beta_i\, r_{M,t} + \varepsilon_{i,t}$$

If we assume that residuals are uncorrelated across stocks then

$$\mathrm{Cov}(r_{i,t},\, r_{j,t}) \;=\; \beta_i\, \beta_j\, \sigma_M^2 \quad (i \neq j)\,, \qquad \mathrm{Var}(r_{i,t}) \;=\; \beta_i^2\, \sigma_M^2 + \sigma_{\varepsilon_i}^2$$

Comparison. Assume 100 stocks:

- Historical covariance matrix: has $100 + \frac{1}{2} \times 100 \times 99 = 5{,}050$ parameters.
- CAPM method: for each stock run a regression and get α, β and residual variance, i.e. 300 required parameters. Plus the variance of the return on the market: 301 parameters in total.

Factor model: structure III

Properties and implications:

- Factors are ex-ante unknown: their identities must be discovered via some statistical / economic analysis.
- Factor exposures/loadings could be time-varying (although I have not given them a time-subscript above):
  - Rolling window OLS
  - Kalman Filter
- If they vary, factor loadings are often treated as known at the beginning of period t.
- We could write down a factor model for excess returns rather than raw returns.

The use of a factor model for risk analysis I

First, let us rewrite the factor model in vector form:

$$\mathbf{R}_t \;=\; \boldsymbol{\alpha} + \mathbf{B}\,\mathbf{F}_t + \boldsymbol{\varepsilon}_t \tag{2}$$

where $\mathbf{R}_t$, $\boldsymbol{\alpha}$ and $\boldsymbol{\varepsilon}_t$ are $N \times 1$ vectors, $\mathbf{B}$ is an $N \times K$ matrix of factor loadings and $\mathbf{F}_t$ is a $K \times 1$ vector of factor returns.
Via this expression, we can write the return covariance matrix, $\mathbf{C}_t$, as:

$$\mathbf{C}_t \;=\; \mathbf{B}\,\boldsymbol{\Sigma}_t\,\mathbf{B}' + \boldsymbol{\Delta}_t \tag{3}$$

where $\boldsymbol{\Sigma}_t$ is the $K \times K$ matrix of factor return covariances and $\boldsymbol{\Delta}_t$ is the diagonal matrix containing the variances of the idiosyncratic risk terms.

The use of a factor model for risk analysis II

Parameter counts: the covariance matrix will be the core of any risk estimate (be it portfolio return variance or value at risk, for example).

- If estimated via the sample covariance, it contains $\frac{1}{2}N(N+1)$ distinct elements.
- If estimated using the factor model, it rests on the estimation of $N(K+1)+\frac{1}{2}K(K+1)$ parameters.

In typical equity applications, N is much larger than K, implying that the sample covariance requires estimation of a lot more parameters, due to the extra structure that the factor model places on the data generating process.

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The use of a factor model for risk analysis III

Practical risk modelling. If we had already estimated factor loadings and factor returns, we would use the following process to derive the return covariance matrix:

1. Use factor returns to estimate the factor return covariance matrix, $\boldsymbol{\Sigma}_t$.
2. Estimate idiosyncratic returns using equation (2).
3. Use the idiosyncratic returns from step 2 to estimate $\boldsymbol{\Delta}_t$.
4. Finally, plug $\boldsymbol{\Delta}_t$ and $\boldsymbol{\Sigma}_t$ into equation (3), along with factor loadings, to derive $\mathbf{C}_t$.

Example 1: covariance matrix for MSCI country data

Data: monthly returns on iShares MSCI ETFs for the UK and Japan. Data cover Feb 2010 to Apr 2026. Also return on the MSCI World.

CAPM, using the MSCI World as the market and country indices as assets:

Country	Beta	Residual Risk
UK	0.94	2.24%
Japan	0.74	2.63%

Using the data we see that:

- (Monthly) market return variance is 0.00177
- CAPM-implied covariance $= 0.94 \times 0.74 \times 0.00177 = 0.00123$
- Sample covariance between UK and Japan returns $= 0.00127$

Example 2: MSCI Barra global equity model

A standard factor model for global equity portfolios:

- Includes a world factor, intended to capture broad global market movement across the model universe
- Includes country factors, capturing the portion of returns associated with national markets
- Includes industry factors, which capture common patterns across companies in the same industry
- Includes style factors, such as momentum, volatility, value, size, size non-linearity, growth, liquidity, and financial leverage
- Includes currency factors, which measure the sensitivity of securities to moves in different currencies

Alternative risk measures for portfolio analysis I

As discussed in the previous lecture, portfolio return variance is by no means the only way to measure risk. Other common measures used in portfolio construction and evaluation include:

- Value at Risk (VaR)
    - The value of portfolio return for which there is a 1% chance of obtaining an outcome below a threshold:
    $$\Pr(R_p < -\mathrm{VaR}_{0.01}) = 0.01$$
    - Relies on knowledge of the tail of the return distribution.
    - If willing to assume that returns are multivariate normal, then it is easier to estimate. But there is lots of evidence that this assumption is false, especially at high frequency.

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Alternative risk measures for portfolio analysis II

- Downside semi-variance
  - Construct a variance measure, but only for observations below the mean:

$$\sigma_d^2 = E\left[(R-\mu)^2 \, \mathbb{1}\{R<\mu\}\right]$$

  - Advantage: focuses on the side of the distribution that is most painful for investors.

- Maximum drawdown
  - Defined as the largest peak-to-trough decline in a portfolio value (not return) series.
  - Used a lot in practice by PMs as a measure of the largest amount they could lose by following a particular signal.
  - Being a 'max' figure, it is not clear what the distribution of this statistic is.

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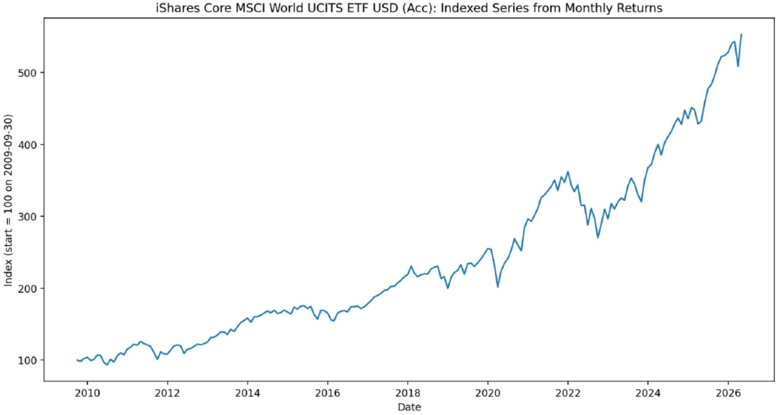
MSCI World ETF cumulative returns



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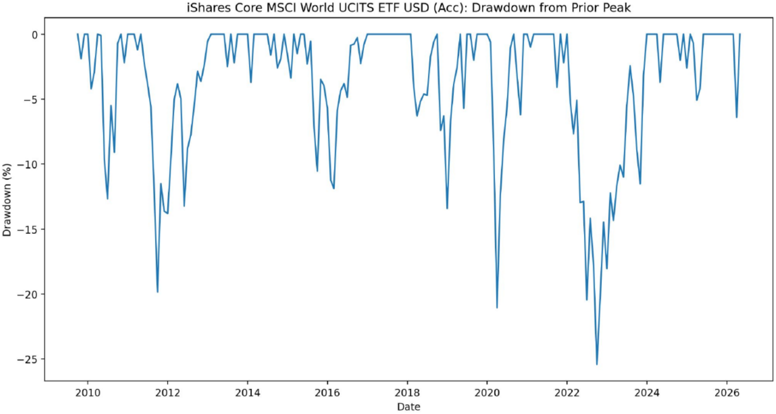
MSCI World ETF maximum drawdowns



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Summary

- Discussed the determinants of IRs:

$$\text{IR} \;=\; \text{IC} \times \sqrt{\text{BR}}$$

- Spent time talking about return forecasting. We argued that alphas should be estimated as

$$\widehat{\alpha} \;=\; \text{IC} \cdot \sigma_{\text{resid}} \cdot z_{\text{signal}}$$

- Discussed factor risk models and how they are used to estimate variances and covariances:

$$\mathbf{C}_t \;=\; \mathbf{B}\,\mathbf{\Sigma}_t\,\mathbf{B}' + \mathbf{\Delta}_t$$

- Saw how we can infer the market-implied returns based on a reverse-engineered version of the mean-variance optimisation.